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SOLVED BY: WAQAR HUSSAIN **PRESENTED BY: TALEEMI MARKAZ**

Q.No.1: Define Urysohn's Lemme. **Marks = 02**

ANSWER: Let X be a normal space; A and B be two disjoint closed subsets of X. Let $[a, b]$ be a closed interval in $\mathbb{R}$. Then there exists a continuous map

$$f: X \to [a, b]$$

such that $f(x) = a \; \forall \; x \in A$ and $f(x) = b \; \forall \; x \in B$.

Q.No.2: Define connected space. **Marks = 02**

ANSWER: A topological space (X, T) is said to be connected if it cannot be written as union of two nonempty open disjoint subsets U and V of X.

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Q.No.3: Define fixed point of a map? Marks = 02

ANSWER: A point $x \in X$ is said to be a fixed point of a map $f: X \to X$ iff $f(x) = x$.

Example: 1- $f: \mathbb{R} \to \mathbb{R}$, defined as $f(x) = x^2$. Here 0 is a fixed point of f. 1 is also a fixed point of f.

2- $g: \mathbb{R} \to \mathbb{R}$, defined as $g(x) = x + 1$. Here g has no fixed point.

Q.No.4: Define lindelof space? Marks = 02

ANSWER: A topological space (X, T) is called a Lindelof space if and only if every open cover of X has a countable subcover.

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Q.No.5: Show that T_1 is T_0 −Space. **Marks = 03**

Given: X is T_1 −Space.

To Prove: X is T_0 −Space.

Proof: Let X is T_1 −Space.

By definition of T_1 −Space $x, y \in X$ such that $x \neq y$. There exists open sets G and H such that

$$x \in G \text{ and } x \notin H \quad \ldots\ldots (1)$$

$$y \in H \text{ and } y \notin G \quad \ldots\ldots (2)$$

Now T_0 −Space: A space where $x, y \in X$. There exist an open set G such that $x \in G$ and $y \notin G$. It is T_0 − Space.

Condition of T_0 − Space holds from eq.(1) and eq.(2). Hence it proved that every T_1 −Space is T_1 −Space.

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Q.No.6: Consider $X = \{a, b, c\}$ with $\tau = \{\emptyset, \{a\}, \{b, c\}, X\}$. (X, τ) is a regular space. Marks = 03

Solution: Let Open Sets are $= \emptyset, \{a\}, \{b, c\}, X$

Let Closed Sets are $= X, \{b, c\}, \{a\}, \emptyset$

Let $a \in X$

$a \in \{a\}$ (open)

And $\{a\} \cap \{b, c\} = \emptyset$

So it is regular space.

Also $a \in \{b, c\}$ (close)

$\{b, c\}_{close} \subseteq \{b, c\}_{open}$

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Q.No.7: Consider $(\mathbb{R}, T_u)$ then $A = (1,3)$ and $B = [3,5)$ are not separated sets. Marks = 03

Solution: $A = (1,3)$ and $B = [3,5)$

Now $\overline{A} = [1,3]$ and $\overline{B} = [3,5]$

Now $A \cap \overline{B} = (1,3) \cap [3,5]$

$A \cap \overline{B} = \emptyset$

Now $\overline{A} \cap B = [1,3] \cap [3,5)$

$A \cap \overline{B} = [3] \neq \emptyset$

So it is not separated.

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Q.No.8: Consider $X = \{a, b, c, d, e\}$ with

$$\tau = \{\emptyset, \{a, b\}, \{d, e\}, \{a, b, c\}, \{a, b, d, e\}, X\}.$$

Then X is a disconnected space. **Marks = 03**

Solution: $X = \{a, b, c, d, e\}$ and

$\tau = \{\emptyset, \{a, b\}, \{d, e\}, \{a, b, c\}, \{a, b, d, e\}, X\}$

Now we take two subsets $\{d, e\}$ and $\{a, b, c\}$

Now $\{d, e\} \cap \{a, b, c\} = \emptyset$

$\{d, e\} \cup \{a, b, c\} = \{a, b, c, d, e\} = X$

So, X is a disconnected space.

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Q.No.9: Let (X, τ) be a compact space and (X, τ') be a Hausdorff space. Then a continuous map $f: (X, \tau) \to (Y, \tau')$ is a close map.

Marks = 05

Solution: Let A be a close subset of X. It is given that X is compact and we know that a closed subset of a compact space is compact.
So, A is compact. Now f is continuous and we know that image of a compact space under a continuous map is compact. So, $f(A)$ is compact subspace of Y.
Now Y is Hausdorff and it has been proved that every compact subspace of a Hausdorff is closed, so $f(A)$ is closed.
Hence it proved that f is a closed map.

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Q.No.10: Consider $(\mathbb{R}, \tau_u)$ and show that there exists no homomorphism between $(0,2)$ and $(0,1] \cup [3,4)$. Marks = 05

Solution: Let $(0,2) \to (0,1] \cup [3,4)$

$\therefore f$ is continuous with respect to τ_u on $\mathbb{R}$.

$(0,2)$ is connected, so

$f(A)$ is also connected

But $(0,1] \cup [3,4)$ is not connected

So, $f(A) \to (0,1]$ or $[3,4)$

There exist no bijective map f.

Hence it is proved that

$$(0,2) \not\approx (0,1] \cup [3,4).$$

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Q.No.11: Show that a set X with cofinite topology τ_{cof} is compact.

Marks = 05

Proof: If X is finite. Then $\tau_{cof} = P(X) \Longrightarrow (X, \tau_{cof})$ is compact.

But if X is not finite. Then we have to show this is a compact space. For this we have to take an arbitrary open cover D and show that there exist a finite sub cover C of that open cover.

Consider an open cover D of X.

Now Let $D \in D$ $\therefore D^c$ is finite.

Now for evry $x \in D^c$. Choose an element of D, D_x containing x. Since

$\{D_x | x \in D^c\}$ is finite $\therefore D^c$ is finite

Consider this subcover C of D. $C = \{D/U\{D_x | x \in D^c\}\}$.

$\Longrightarrow C$ is a finite subcover of D. $\Longrightarrow (X, \tau_{cof})$ is compact.

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Q.No.12: Show that every cofinite space is T_1. Marks = 05

Proof: Let (X, τ_{cof})

Let X is finite. So

$\tau_{cof} = P(X)$ (Discrete Topology)

Let X is not finite. So

$U \in \tau_{cof}$ iff

$$U = \emptyset \text{ or } U^c = finite$$

So every finite subset of closed finite is closed

$x, y \in X$ with cofinite topology

$$U_x = X/\{y\} \text{ and } U_y = X/\{x\}$$

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